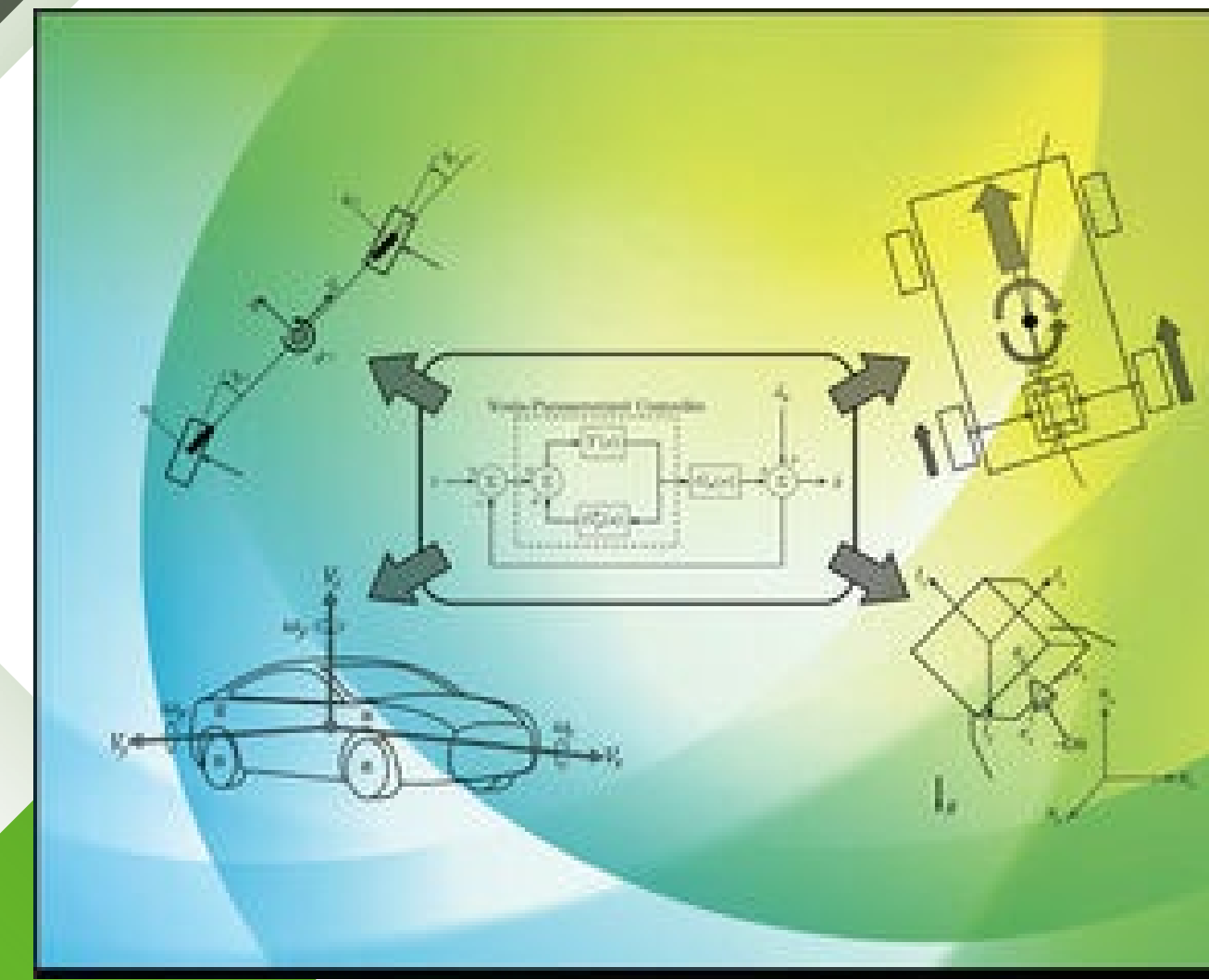


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# Introduction to Robust Control (Part 2)



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xxxx



## Remember

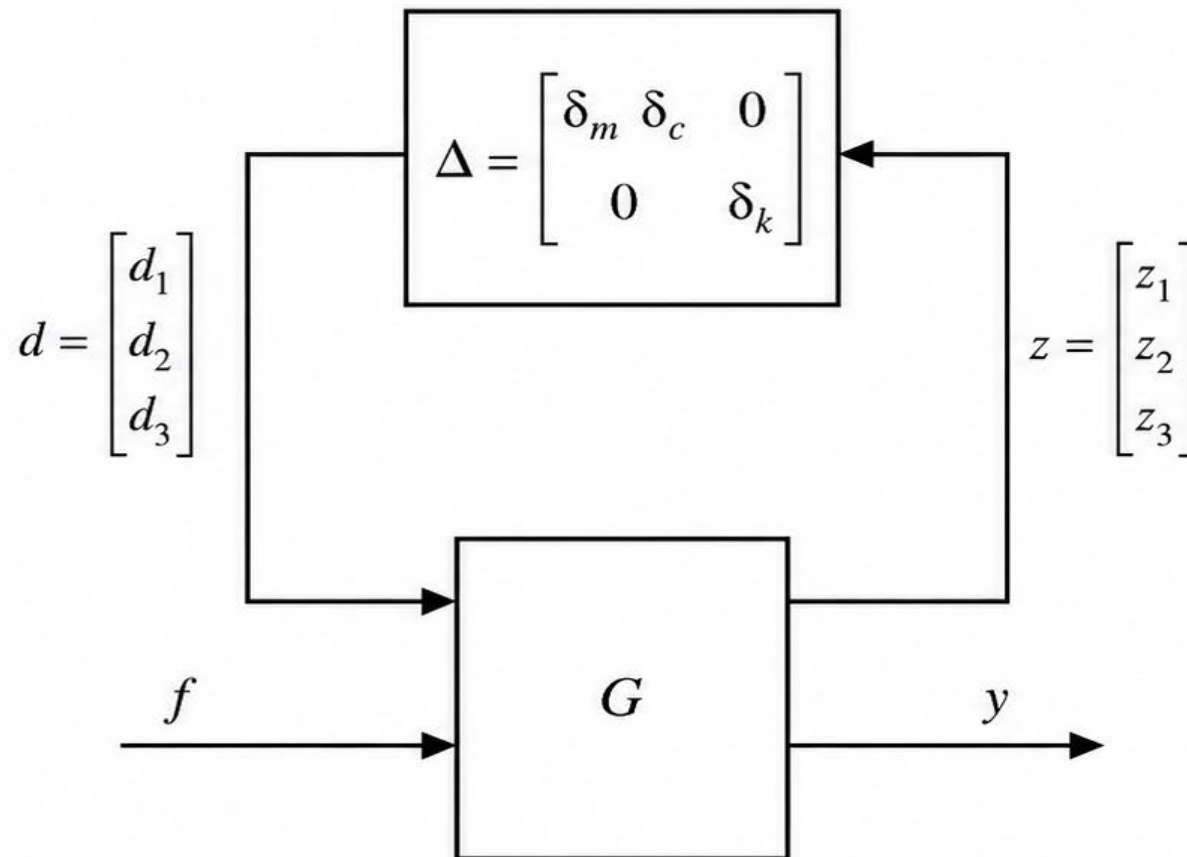
- **Robust control theory. No mathematical model accurately represents the physical system.**
- **The real  $G(s)$  may vary due to manufacturing tolerances, changes in mass, temperature, aging, unmodeled dynamics, delays, identification errors, etc.**
- **Robust control works with a family of possible plant models.**



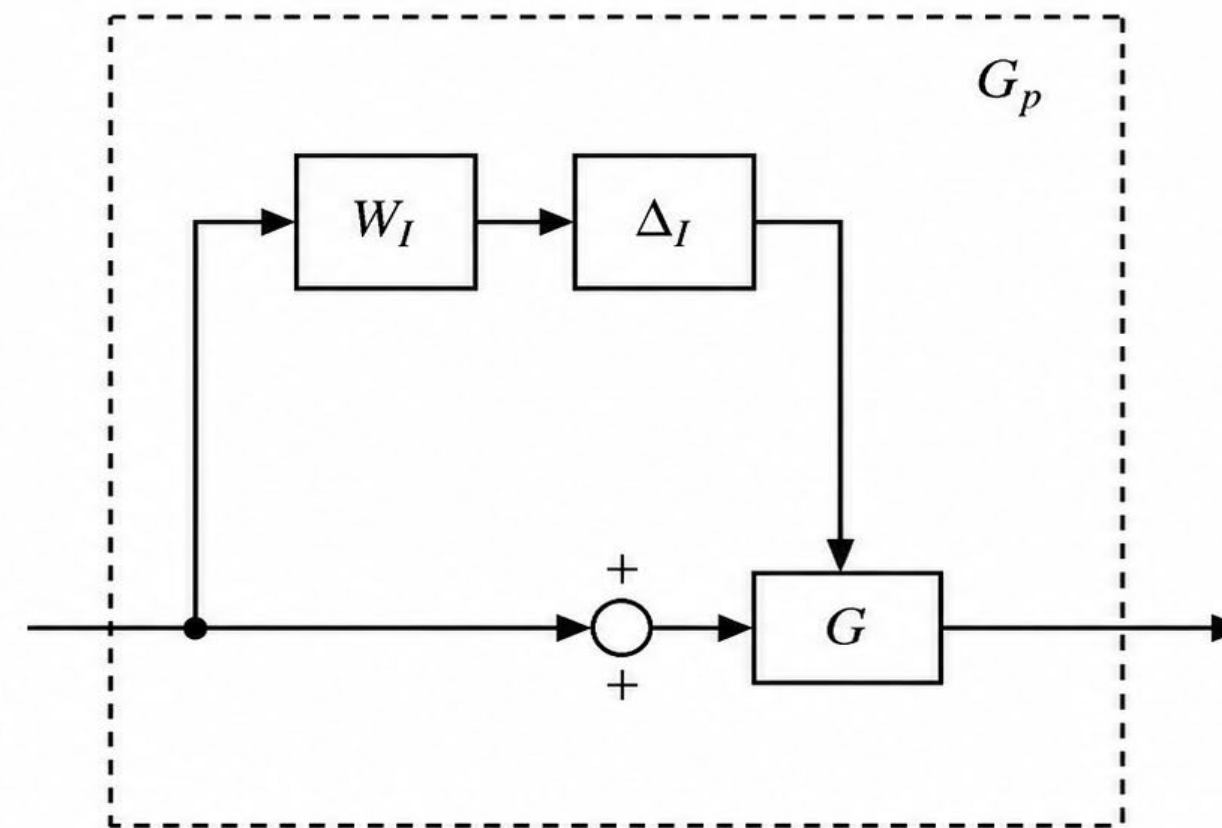
# Remember

Difference between structured and unstructured uncertainty lies in how much information we preserve about how the plant can vary.

Structured uncertainty



Unstructured uncertainty



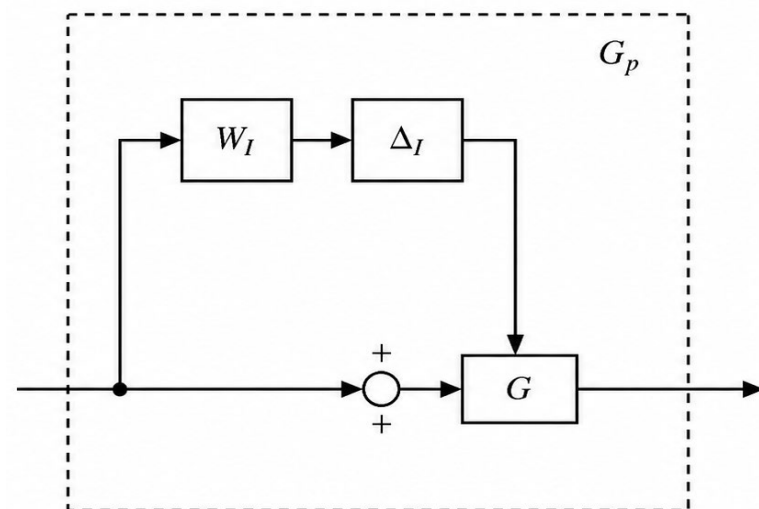


# Remember

**Unstructured uncertainty groups many modeling errors into a single block  $\Delta$ .**

## Note 1:

Unstructured uncertainty does not mean a complete lack of knowledge about the plant. It means that we do not know—or do not wish to preserve—the exact internal cause of the error or uncertainty, but we do have sufficient information to constrain its effect on the input–output relationship.



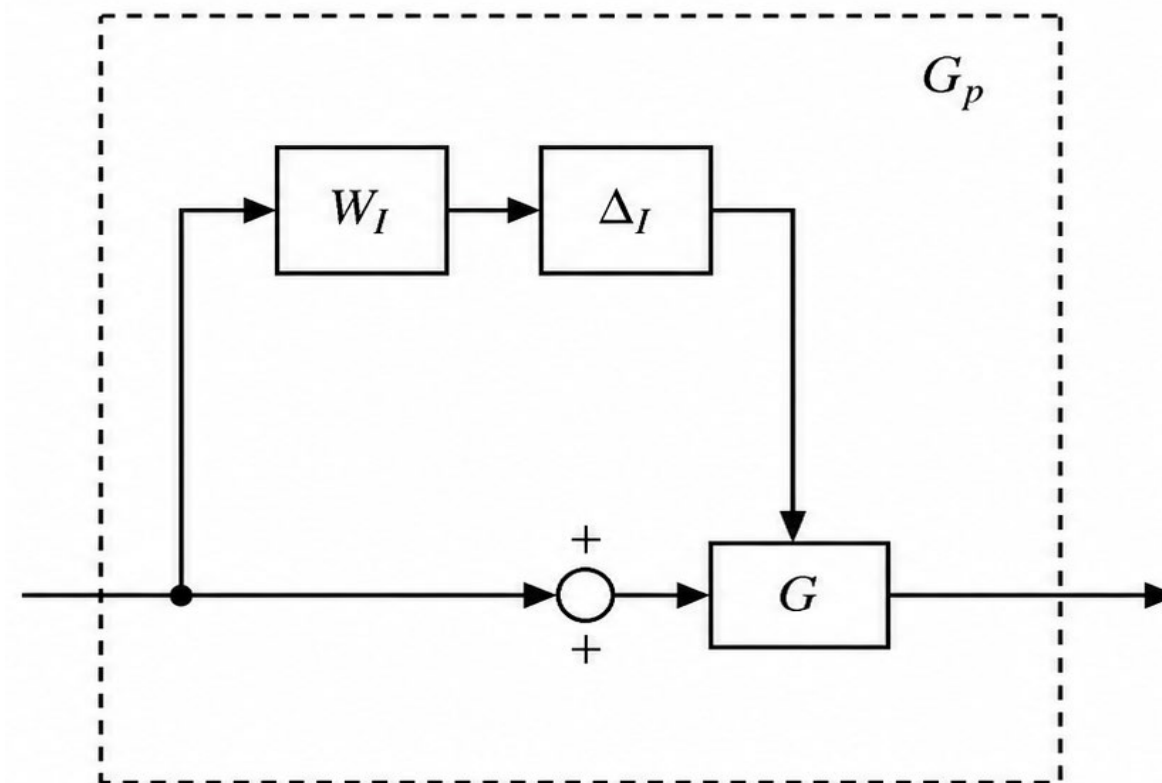
$$\|\Delta\|_{\infty} < 1$$



# Remember

## Note 2:

- Unstructured uncertainty. We don't know exactly what changes inside the plant, but I can bound how much its response changes.
- Delta is stable and satisfies  $\|\Delta\|_{\infty} < 1$ .
- Unstructured means that uncertainty or error does not increase to certain magnitude in each frequency.



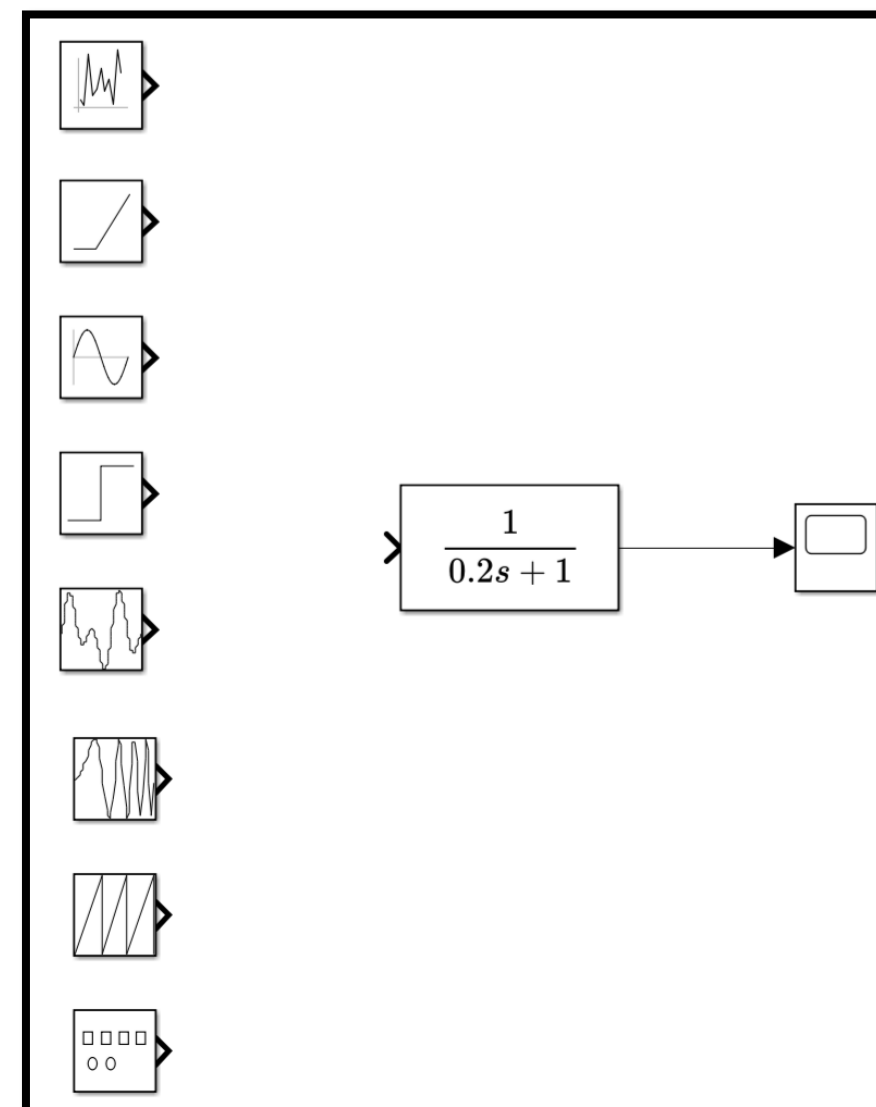
$$\|\Delta\|_{\infty} < 1$$





## Example 1:

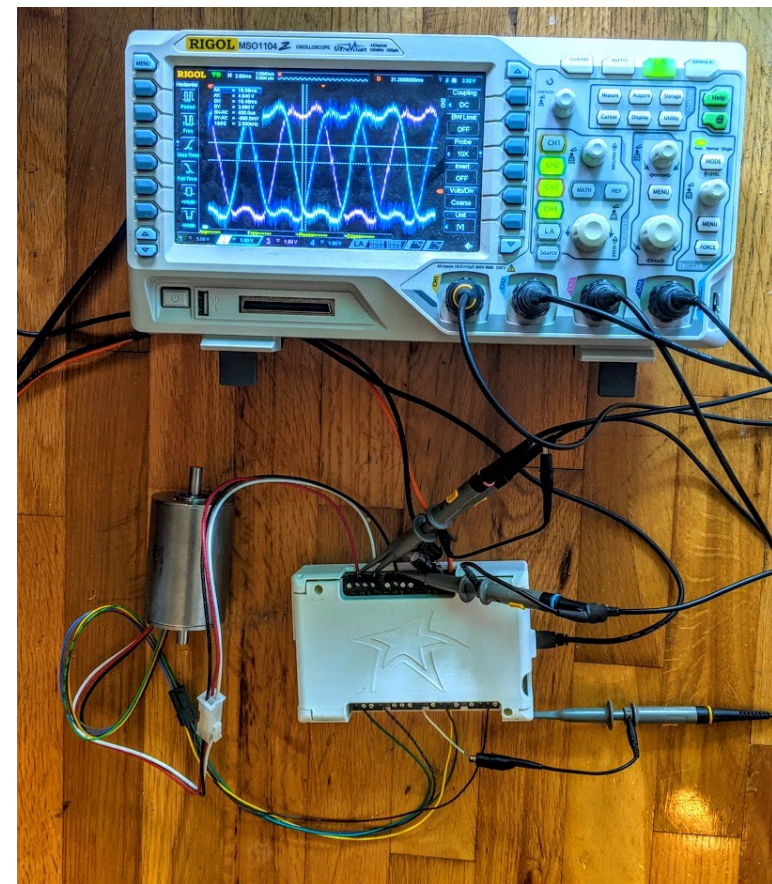
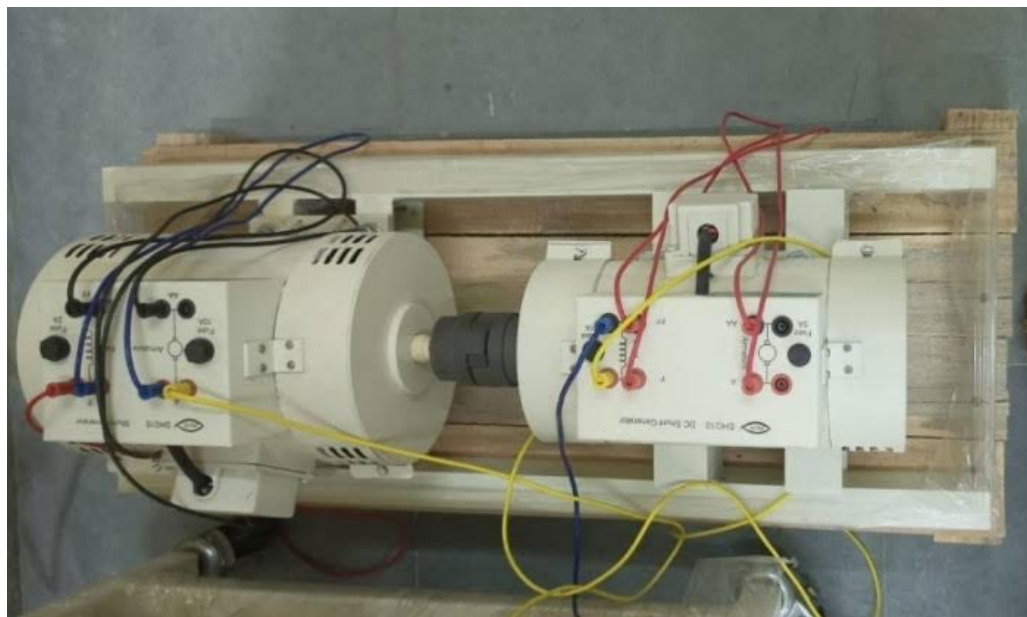
Assume that the nominal model of a motor is its transfer function. We can program this transfer function into software and obtain a complete analysis of its time- and frequency-domain responses.





# Remember

In the laboratory, using a real system, we can apply known signals to the plant, measure its response, and compare it with the response of the nominal model.



| Experiment | Temperature | Load | Operating point | Response         |
|------------|-------------|------|-----------------|------------------|
| 1          | 25 °C       | 0%   | Low velocity    | Step             |
| 2          | 25 °C       | 25 % | Velocidad media | Step             |
| 3          | 25 °C       | 50 % | Nominal         | Sinusoidal Sweep |
| 4          | 40 °C       | 0%   | Low velocity    | Step             |
| 5          | 40 °C       | 50 % | Nominal         | Sinusoidal Sweep |
| 6          | 60 °C       | 75 % | High Velocity   | Señal PRBS       |



# Remember

For multiplicative uncertainty we take the relative error

$$E_i(j\omega) = \frac{G_i(j\omega) - G_0(j\omega)}{G_0(j\omega)}$$

Then, the magnitude of the multiplicative relative error

$$r_i(\omega) = \left| \frac{G_i(j\omega) - G_0(j\omega)}{G_0(j\omega)} \right|$$

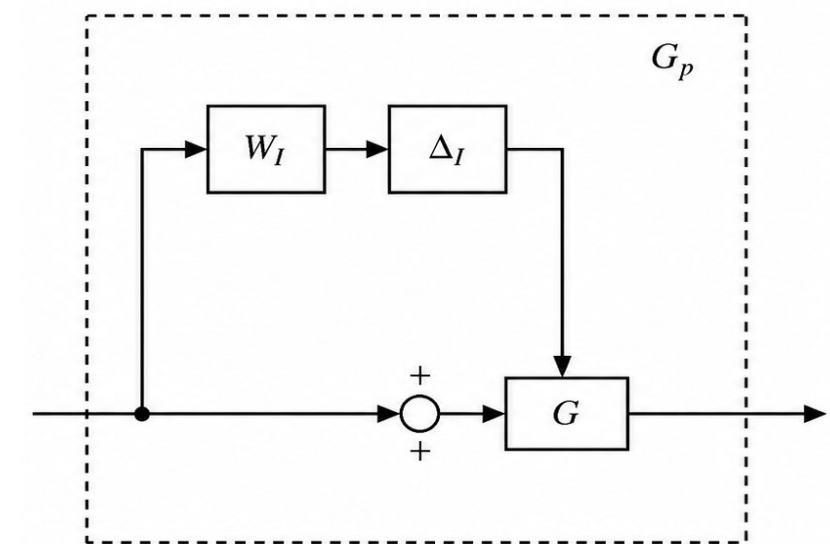
We can take the max() error for the whole range

$$r_{\max}(\omega) = \max_{i=1, \dots, N} r_i(\omega)$$

The design implies fit a stable  $W_m(s)$  that satisfies

$$|W_m(j\omega)| \geq r_{\max}(\omega) \quad \forall \omega$$

$W_m(s)$  a frequency envelope. Therefore,  $W_m(s)$  does not explain the cause of the variation, only capture its magnitude.







# Remember

## Matlab Example

```
C:\Users\betot\iCloudDrive\Documents\03-EL INGE DE CONTROL\Matlab_Work\B_WB01_Unstructured Uncertainty.mlx
```



**Taller: Control Robusto y Estocástico**

**Tema: Incertidumbre No Estructurada**

**Dr. Julio A. García Rodríguez**

**EXERCISE**

G0 = nominal model used to design the controller

G1 to G6 = possible physical plants or experiments

W = Weighting function covering discrepancies

Delta = unknown dynamics

```
1 % Part A: Nominal model
2
3 clear, clc, close all;
4
5 s = tf('s');
6
7 G0 = 1/(0.2*s + 1); %Nominal Servomotor Model
8
```

